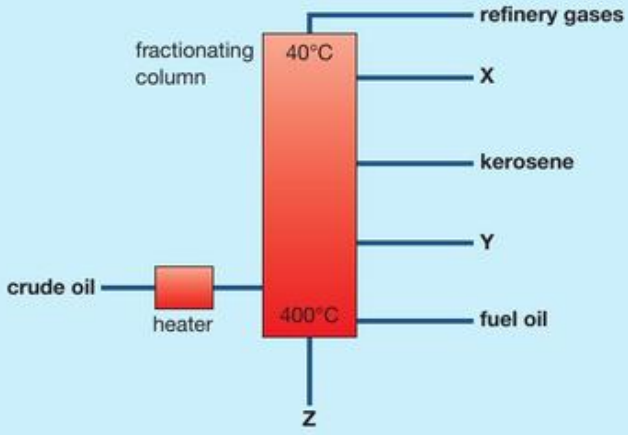
UNIT QUESTIONS

SKILLS ANALYSIS 6

1 Crude oil is a complex mixture of hydrocarbons. The diagram shows the separation of crude oil into simpler mixtures called fractions.



a What could X, Y and Z represent (choose one answer)? **(1)**

	X	Y	Z
A	gasoline	bitumen	diesel
B	diesel	gasoline	bitumen
C	bitumen	gasoline	diesel
D	gasoline	diesel	bitumen

SKILLS CRITICAL THINKING

b State a use for the refinery gas fraction. **(1)**

c Name the liquid that leaves the fractionating column at the lowest temperature. **(1)**

7

d Describe how crude oil is separated into fractions in industry. **(4)**

e Write down and explain the relationship between the number of carbon atoms in a hydrocarbon and its boiling point. **(2)**

SKILLS PROBLEM SOLVING 9

f One of the hydrocarbons, $C_{15}H_{32}$, called pentadecane, is present in the kerosene fraction. It could be used as a fuel in jet engines. Write an equation for the incomplete combustion of pentadecane to produce carbon monoxide. **(2)**

g The complete combustion of alkanes produces carbon dioxide, CO_2, which can dissolve in water to form a weakly acidic solution.

SKILLS INTERPRETATION 8

i Draw a dot-and-cross diagram to show the bonding in CO_2. Show the outer electrons only. **(2)**

SKILLS REASONING 7

ii Predict the most likely pH of a solution of CO_2 (choose one answer). **(1)**

A 1 **B** 5 **C** 7 **D** 9

SKILLS CRITICAL THINKING 6

h $C_{15}H_{32}$ is a member of the alkane family. Long-chain alkanes are often cracked to form alkenes.

i Write down two conditions used for cracking alkanes in industry. **(2)**

ii Give two reasons why cracking is necessary in the oil industry. **(2)**

7

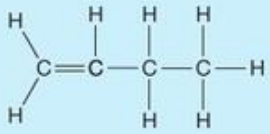
iii During cracking, one molecule of $C_{15}H_{32}$ is converted into one molecule of ethene, one molecule of another alkene C_4H_8 and one molecule of hydrocarbon **W**. State the molecular formula of **W**. **(1)**

6

iv Alkenes are unsaturated hydrocarbons. Write down what is meant by the term *unsaturated*. **(1)**

SKILLS INTERPRETATION 8

v One possible structure for C_4H_8 is

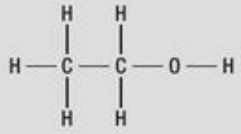
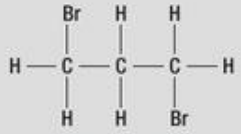


Draw a displayed formula for each of the other two isomers of C_4H_8, which are also alkenes. **(2)**

(Total 22 marks)

SKILLS ANALYSIS 6

2 The table shows the formulae of some organic compounds.

A	B	C
$CH_2=CHCl$	$CH_3CH_2CH_3$	$CH_2=CHCH_2CH_3$
D	**E**	**F**
C_2H_6		

a Give one reason why compound **E** is *not* a hydrocarbon. **(1)**

b **i** State the name of compound **F**. **(1)**

SKILLS INTERPRETATION 8

ii Draw the displayed formula of an isomer of compound **F**. **(2)**

SKILLS PROBLEM SOLVING

c Compound **C** can undergo an addition reaction with bromine. Complete the equation below showing the structural formula of the product(s). **(1)**

$CH_2=CHCH_2CH_3 + Br_2 \rightarrow$

d Two of the compounds shown belong to the same homologous series.

SKILLS CRITICAL THINKING 6

i Give the letters of these two compounds. **(1)**

ii Write down the general formula of this homologous series. **(1)**

iii Write down *two* other characteristics of the compounds in a homologous series. **(2)**

e Compound **A** can undergo addition polymerisation.

i Give the name of the addition polymer formed by compound **A**. **(1)**

7

ii Draw the repeat unit of the addition polymer of compound **A**. **(1)**

SKILLS REASONING 7 6

iii Explain the problems in the disposal of the addition polymer formed by compound **A**. **(2)**

iv Write down and explain the observations you would make if bromine water was added to the addition polymer of **A**. **(2)**

SKILLS CRITICAL THINKING 5 6

f Compound **B** undergoes a reaction with chlorine.

i Write down the condition required for this reaction to take place. **(1)**

ii Write down the type of reaction that occurs. **(1)**

SKILLS INTERPRETATION 8

iii Draw the displayed formulae for the two possible organic products from the reaction between compound **B** and chlorine. **(2)**

(Total 19 marks)

SKILLS PROBLEM SOLVING 9

3 Compound **X** contains 55% carbon, 9% hydrogen and 36% oxygen by mass.

a Calculate the empirical formula of **X**. **(3)**

b The relative molecular mass (M_r) of **X** is 88. Deduce the molecular formula of **X**. **(1)**

CHEMISTRY ONLY

SKILLS INTERPRETATION 8

c **X** belongs to a class of compound called the carboxylic acids. Draw the displayed formula of **X** and name the compound. **(2)**

SKILLS PROBLEM SOLVING 9

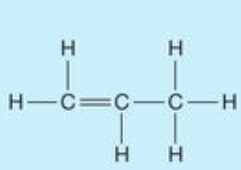
d Write a balanced equation for the reaction between **X** and methanol. Name the organic product formed. **(3)**

END OF CHEMISTRY ONLY

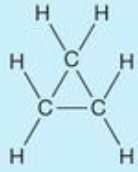
(Total 9 marks)

SKILLS EXECUTIVE FUNCTION 6

4 Below are two hydrocarbons with the molecular formula C_3H_6.



compound 1



compound 2

a Describe a chemical test for distinguishing between the two hydrocarbons and give the results for this test. **(3)**

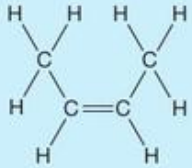
Test:

Result with compound 1:

Result with compound 2:

SKILLS CRITICAL THINKING 7

b Compound 3 is in the same homologous series as compound 1. Compound 3 can form an addition polymer. Draw the repeat unit of this polymer. **(2)**



compound 3

SKILLS CRITICAL THINKING 6

c Name the addition polymer formed by compound 3. **(1)**

(Total 6 marks)

CHEMISTRY ONLY

SKILLS INTERPRETATION 7 **5**

Esters occur naturally, but can be made in the laboratory from an alcohol and a carboxylic acid. Esters have many uses.

a Butyl ethanoate is an important industrial solvent.

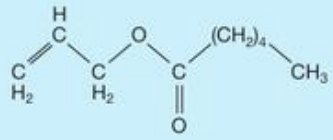
i Draw the displayed formula of butyl ethanoate. **(2)**

SKILLS CRITICAL THINKING 5

ii Butyl ethanoate can be made from butanol and ethanoic acid. Give the conditions for making ethanoic acid from ethanol in a lab. **(3)**

SKILLS INTERPRETATION 8

b Esters can be used in perfumes. One of them, allyl hexanoate, is used because it smells of pineapple. Draw out the structural formulae of the alcohol and the carboxylic acid used to make allyl hexanoate. **(2)**



SKILLS CRITICAL THINKING

c Imagine that you wish to prepare ethyl ethanoate according to the reaction:

$CH_3COOH(l) + CH_3CH_2OH(l) \rightleftharpoons CH_3COOCH_2CH_3(l) + H_2O(l)$

6

i Write down the meaning of the sign $\rightleftharpoons$. **(1)**

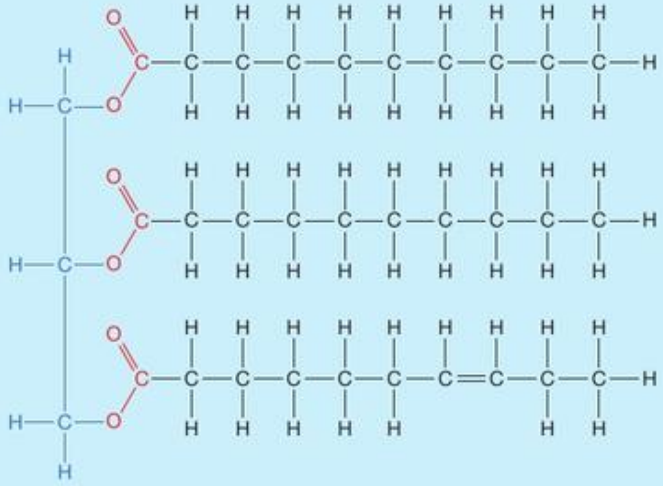
ii The reaction is catalysed in order to increase the rate of the reaction. State the name of the catalyst used for this reaction in a laboratory. **(1)**

7

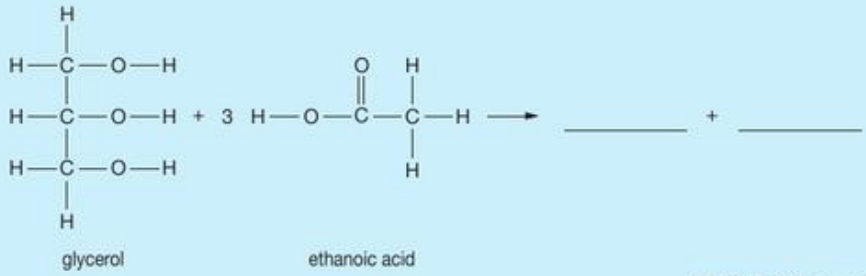
d Write down and explain the effect of the catalyst on the position of equilibrium in the reaction in **c**. **(2)**

SKILLS PROBLEM SOLVING 8

e Fats are naturally occurring esters. The diagram below shows one type of fat in our body, a triglyceride.



Triglycerides are made from the alcohol glycerol and different long-chain carboxylic acids. The structure of glycerol is shown below. Complete and balance the equation for the reaction between glycerol and ethanoic acid. **(3)**



(Total 14 marks)

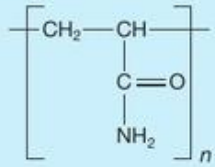
END OF CHEMISTRY ONLY

SKILLS INTERPRETATION 8

6 This question is about synthetic polymers. Synthetic polymers can be either addition polymers or condensation polymers.

a Poly(propene) is an addition polymer made from propene $CH_2=CHCH_3$. Draw a length of the poly(propene) chain showing two repeat units. **(2)**

b Another addition polymer, polyacrylamide, is used to make hydrogel in contact lenses, nappies and drug delivery systems. The structure of the polymer is shown in the diagram. Draw the structural formula of the monomer used to make polyacrylamide.



(1)

CHEMISTRY ONLY

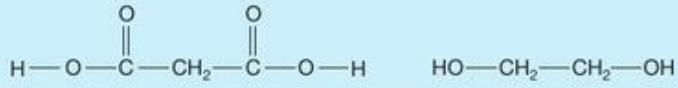
c Polyesters are condensation polymers.

SKILLS CRITICAL THINKING 7

i Describe how condensation polymerisation is different from addition polymerisation. **(2)**

SKILLS INTERPRETATION 8

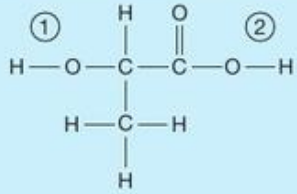
ii The structures of two monomers that are used to make a polyester are:



Draw the structure of the repeat unit of the polyester formed from these two monomers. **(2)**

SKILLS CRITICAL THINKING

d Polylactic acid is a biopolyester. Lactic acid has the structure shown below.



7

i Explain the meaning of *biopolyester*. **(1)**

SKILLS CRITICAL THINKING 5

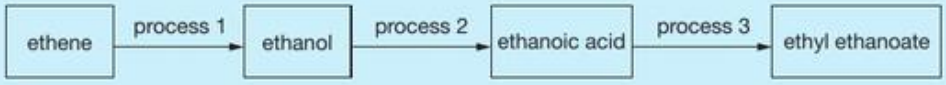
ii Which of the two –OH groups labelled can react with aqueous sodium hydroxide (choose one answer)? **(1)**

	1	2
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

(Total 9 marks)

SKILLS ANALYSIS, PROBLEM SOLVING

7 The diagram shows one possible synthetic route to make ethyl ethanoate, an important solvent used in glues and nail polish removers.

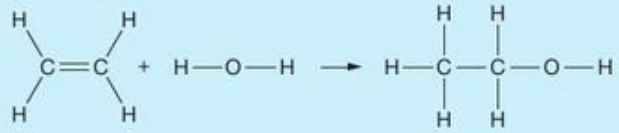


a Process 1 is called catalytic hydration.

i Name the catalyst and state one more condition required for this reaction in industry. **(2)**

10

ii The equation for the conversion of ethene to ethanol can be written using displayed formulae.



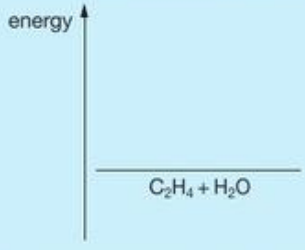
The table below gives some average bond energies.

Bond	C–C	C=C	C–H	C–O	O–H
Average bond energy/kJ/mol	348	612	412	360	463

Use the information in the table to calculate the enthalpy change for this reaction. **(3)**

SKILLS INTERPRETATION 9

iii Complete and label the energy level diagram for this reaction. Show clearly the enthalpy change, ΔH, of the reaction. **(2)**



b Ethanol can also be made from plants, including sugar cane. Yeast is added to a sugar solution and left in the warm for several days in anaerobic conditions.

SKILLS CRITICAL THINKING 5

i Name the process in which ethanol is produced from glucose. **(1)**

6

ii What is the purpose of the yeast? **(1)**

SKILLS REASONING 7

iii Explain why anaerobic conditions are required for this reaction to occur. **(1)**

SKILLS REASONING 6

iv Why do countries such as Brazil use this process to produce ethanol rather than the catalytic hydration of ethene? **(2)**

c In process 2, ethanol is oxidised to form ethanoic acid using potassium dichromate(VI).

SKILLS CRITICAL THINKING 5

i Write down two other conditions required for this reaction to occur in a laboratory. **(2)**

SKILLS PROBLEM SOLVING 9

ii Write a balanced chemical equation for this reaction. (You may use [O] to represent the oxidising agent.) **(2)**

SKILLS CRITICAL THINKING 5

iii What colour change do you observe in this reaction? **(2)**

SKILLS PROBLEM SOLVING 9

iv Ethanoic acid can react with metals to produce hydrogen gas. Write a balanced equation for the reaction between ethanoic acid and magnesium. **(2)**

d In process 3, ethanoic acid reacts with ethanol to form ethyl ethanoate in a condensation reaction.

SKILLS CRITICAL THINKING 6

i Explain why this is called a condensation reaction. **(1)**

5

ii How could you separate the product from the reaction mixture? **(2)**

(Total 23 marks)

END OF CHEMISTRY ONLY

APPENDIX A: PERIODIC TABLE

Period	1	2											3	4	5	6	7	0
1	1 H Hydrogen 1																	4 He Helium 2
2	7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
3	23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
4	39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
5	85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	(98) Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
6	133 Cs Caesium 55	137 Ba Barium 56	■	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	(209) Po Polonium 84	(210) At Astatine 85	(222) Rn Radon 86
7	(223) Fr Francium 87	(226) Ra Radium 88	■	(267) Rf Rutherfordium 104	(268) Db Dubnium 105	(269) Sg Seaborgium 106	(270) Bh Bohrium 107	(277) Hs Hassium 108	(278) Mt Meitnerium 109	(281) Ds Darmstadtium 110	(282) Rg Roentgenium 111	(285) Cn Copernicium 112	(286) Nh Nihonium 113	(289) Fl Flerovium 114	(290) Mc Moscovium 115	(293) Lv Livermorium 116	(294) Ts Tennessine 117	(294) Og Oganesson 118
			■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■

■	139 La Lanthanum 57	140 Ce Cerium 58	141 Pr Praseodymium 59	144 Nd Neodymium 60	(145) Pm Promethium 61	150 Sm Samarium 62	152 Eu Europium 63	157 Gd Gadolinium 64	159 Tb Terbium 65	163 Dy Dysprosium 66	165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69	173 Yb Ytterbium 70	175 Lu Lutetium 71
■	(227) Ac Actinium 89	232 Th Thorium 90	231 Pa Protoactinium 91	238 U Uranium 92	(237) Np Neptunium 93	(244) Pu Plutonium 94	(243) Am Americium 95	(247) Cm Curium 96	(247) Bk Berkelium 97	(251) Cf Californium 98	(252) Es Einsteinium 99	(257) Fm Fermium 100	(258) Md Mendelevium 101	(259) No Nobelium 102	(266) Lr Lawrencium 103

a = relative atomic mass
X = atomic symbol
b = atomic number

(Masses in brackets are the mass numbers of the longest-lived isotope)